«part def»

Nanosatellite EPS Engineering Framework

doc

The system of interest directly realizes configuration management and collaboration and communication. It coordinates externally authored requirements, models, and analysis results through typed interfaces without performing requirements elicitation, model tailoring, or domain analysis. This abstraction does not select a deployment, toolchain, or internal architecture.

«ref part»

systemModel : 'Versioned System Model'

ports

^frameworkControl : FrameworkBoundaryInterfaces::'Model Control Port'

«ref part»

evidenceRepository : 'Evidence Repository'

ports

^frameworkCustody : FrameworkBoundaryInterfaces::'Evidence Custody Port'

«ref part»

analysisTools : 'Analysis Tools'

ports

^frameworkExecution : FrameworkBoundaryInterfaces::'Analysis Execution Port'

missionApplication : FrameworkBoundaryInterfaces::'Mission Application Port'

«perform action»

maintainEvidenceChain : FrameworkBlackBoxFunctions::'Maintain Evidence Chain'

items

^request : 'Evidence Chain Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^registrations : 'Artifact Registration Set'
^scope : 'Evidence Chain Scope'
^assessment : 'Evidence Chain Assessment'

actions

^registerExternalArtifacts : 'Register External Engineering Artifacts'
^scopeEvidenceChain : 'Scope Evidence Chain'
^validateEvidenceChain : 'Validate Evidence Chain'
^issueAssessment : 'Issue Evidence Chain Assessment'

«perform action»

configureFrameworkForCase : FrameworkBlackBoxFunctions::'Configure Framework for EPS Case'

items

^request : 'Case Configuration Request'
^commonCore : 'Common Core Package'
^adaptationDecision : 'Adaptation Decision'
^draft : 'Case Configuration Draft'
^configuration : 'Controlled Case Configuration'
^status : 'Configuration Status'

actions

^captureConfiguration : 'Capture Case Configuration'
^validateConfiguration : 'Validate Case Configuration'
^issueStatus : 'Issue Configuration Status'

«perform action»

reproduceDesignatedAnalysis : FrameworkBlackBoxFunctions::'Reproduce Designated Analysis'

items

^request : 'Reproduction Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^returnedResult : 'Analysis Result Package'
^executionRequest : 'Analysis Execution Request'
^result : 'Analysis Result Package'
^verdict : 'Reproduction Verdict'

actions

^preparePackage : 'Prepare Reproduction Package'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^evaluateResult : 'Evaluate Reproduced Result'
^issueVerdict : 'Issue Reproduction Verdict'

researchGovernance : FrameworkBoundaryInterfaces::'Research Governance Port'

«perform action»

supportEarlyDesignReview : FrameworkBlackBoxFunctions::'Support Early Design Review'

items

^request : 'Design Review Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^findings : 'Review Findings'
^disposition : 'Finding Disposition'
^outcome : 'Review Outcome'
^reviewPackage : 'Review Package'
^issuedFindings : 'Review Findings'
^reproduction : 'Reproduction Verdict'
^status : 'Review Status'

actions

^assemblePackage : 'Assemble Review Package'
^coordinateReproduction : 'Coordinate Selected Reproduction'
^manageFindings : 'Manage Review Findings'
^issueStatus : 'Issue Review Status'

«perform action»

assessControlledDesignChange : FrameworkBlackBoxFunctions::'Assess Controlled Design Change'

items

^request : 'Design Change Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^revisions : 'Change Revision Package'
^returnedResult : 'Analysis Result Package'
^executionRequest : 'Analysis Execution Request'
^impact : 'Change Impact Assessment'
^evidencePackage : 'Change Evidence Package'

actions

^validateRequest : 'Validate Change Request'
^analyzeImpact : 'Analyze Change Impact'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^validateEvidence : 'Validate Regenerated Evidence'
^assembleEvidence : 'Assemble Change Evidence Package'

«perform action»

compareCandidateArchitectures : FrameworkBlackBoxFunctions::'Compare Candidate EPS Architectures'

items

^request : 'Candidate Comparison Request'
^protocol : 'Comparison Protocol'
^candidates : 'Candidate Model Package'
^returnedResults : 'Analysis Result Package'
^sensitivityAndUncertainty : 'Sensitivity and Uncertainty Assessment'
^executionRequest : 'Analysis Execution Request'
^result : 'Analysis Result Package'
^comparisonPackage : 'Candidate Comparison Package'

actions

^validateBasis : 'Validate Comparison Basis'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^determineFeasibility : 'Determine Candidate Feasibility'
^compareCandidates : 'Compare Candidate Architectures'
^issueComparison : 'Issue Candidate Comparison'

independentReview : FrameworkBoundaryInterfaces::'Independent Review Port'

«exhibit state»

operationalControl : FrameworkOperationalState::'Framework Operational Control'

states

^<ST-01> Unconfigured
^<ST-02> Configuring
^<ST-03> Controlled Baseline Ready
^<ST-04> Assessing Evidence
^<ST-05> Reproducing Analysis
^<ST-06> Assessing Change
^<ST-07> Comparing Candidates
^<ST-08> Under Review
^<ST-09> Discrepancy Open
^<ST-10> Recovering

state transition

^initializeControl
^<T-01> beginConfiguration
^<T-02> configurationAccepted
^<T-03> configurationDeficient
^<T-04> reconfigureContext
^<T-05> assessEvidenceRequest
^<T-06> reproduceAnalysisRequest
^<T-07> assessChangeRequest
^<T-08> compareCandidatesRequest
^<T-09> reviewRequest
^<T-10> evidenceAccepted
^<T-11> evidenceDeficient
^<T-12> reproductionAccepted
^<T-13> reproductionDeficient
^<T-14> changeAccepted
^<T-15> changeDeficient
^<T-16> comparisonAccepted
^<T-17> comparisonDeficient
^<T-18> reviewAccepted
^<T-19> reviewDeficient
^<T-20> beginControlledRecovery
^<T-21> retryConfiguration
^<T-22> retryReproduction
^<T-23> retryChange
^<T-24> retryComparison
^<T-25> retryReview
^<T-26> recoveryAccepted
^<T-27> recoveryDeficient

modelControl : FrameworkBoundaryInterfaces::'Model Control Port'

analysisExecution : FrameworkBoundaryInterfaces::'Analysis Execution Port'

evidenceCustody : FrameworkBoundaryInterfaces::'Evidence Custody Port'